

The Power of Vaccination

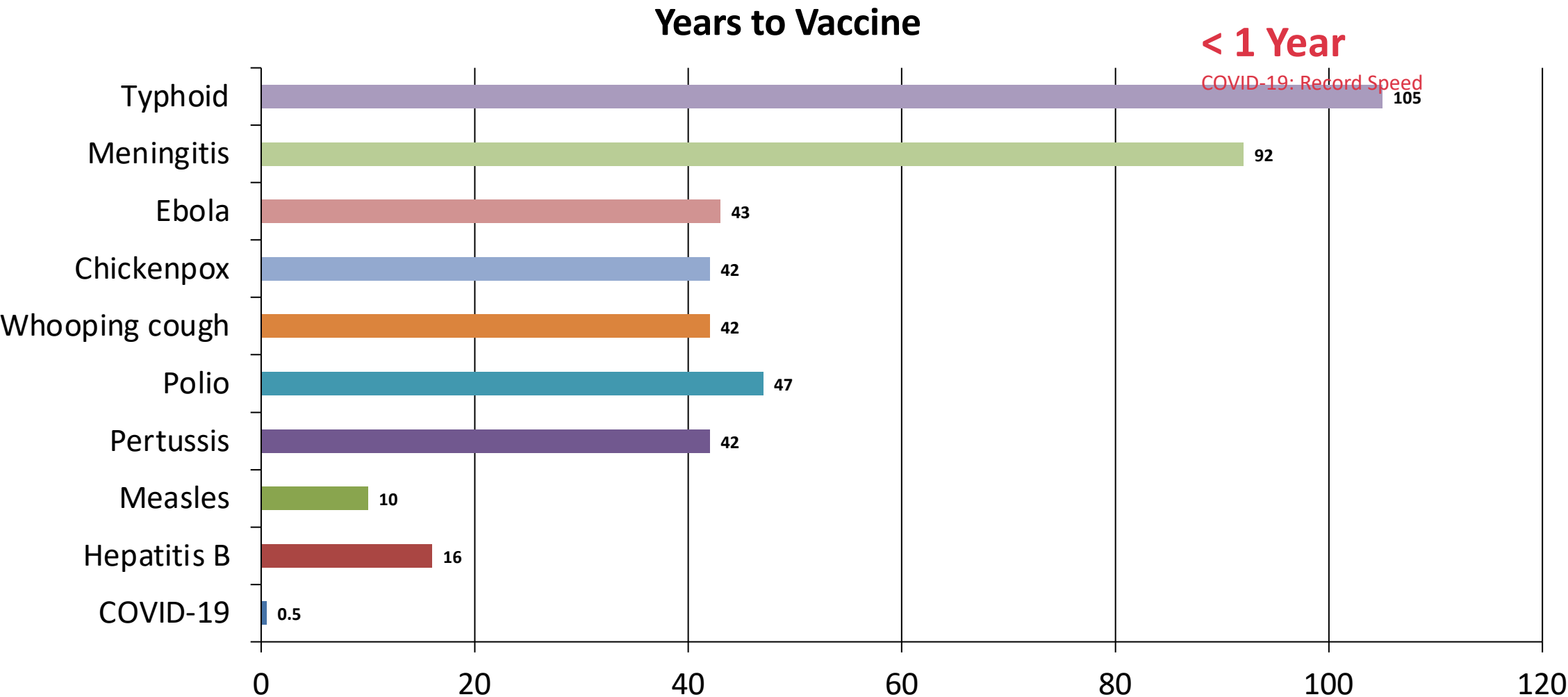
From Smallpox Eradication to COVID-19 in Record Time

A Data-Driven Overview of Vaccine Impact | 1796–2020

Source: Our World in Data | Vaccination

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure (years)



Data: Our World in Data. Typhoid took 105 years; Meningitis 92 years; COVID-19 under 1 year.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases*	Post-Vaccine Cases	Case Reduction	Status
Diphtheria	158	0	100%	ELIMINATED
Smallpox	250	0	100%	ELIMINATED
Acute polio	141	0	100%	ELIMINATED
Measles	3,044	0.2	99.99%	Near Eliminated
Rubella	242	0.04	99.98%	Near Eliminated
Pertussis	1,534	52	96.6%	Significant Reduction
Hepatitis A	465	51	89%	Progress Made
Hepatitis B	273	44	83.9%	Progress Made
Pneumococcal	233	139	40.5%	Gaps Remain

*Per million per year. Data: Roush & Murphy (2007) via Our World in Data.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



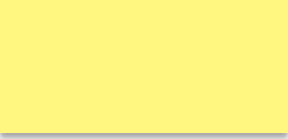
1929-1962: Pre-Vaccine Era

High incidence: 100-1,000+ cases per 100,000 in nearly every state



1963: First Vaccine

Measles vaccine developed by John Enders



1971: MMR Introduced

Combined measles-mumps-rubella vaccine by Maurice Hilleman



1980: Mandate Enacted

Vaccination mandatory for kindergarten entry



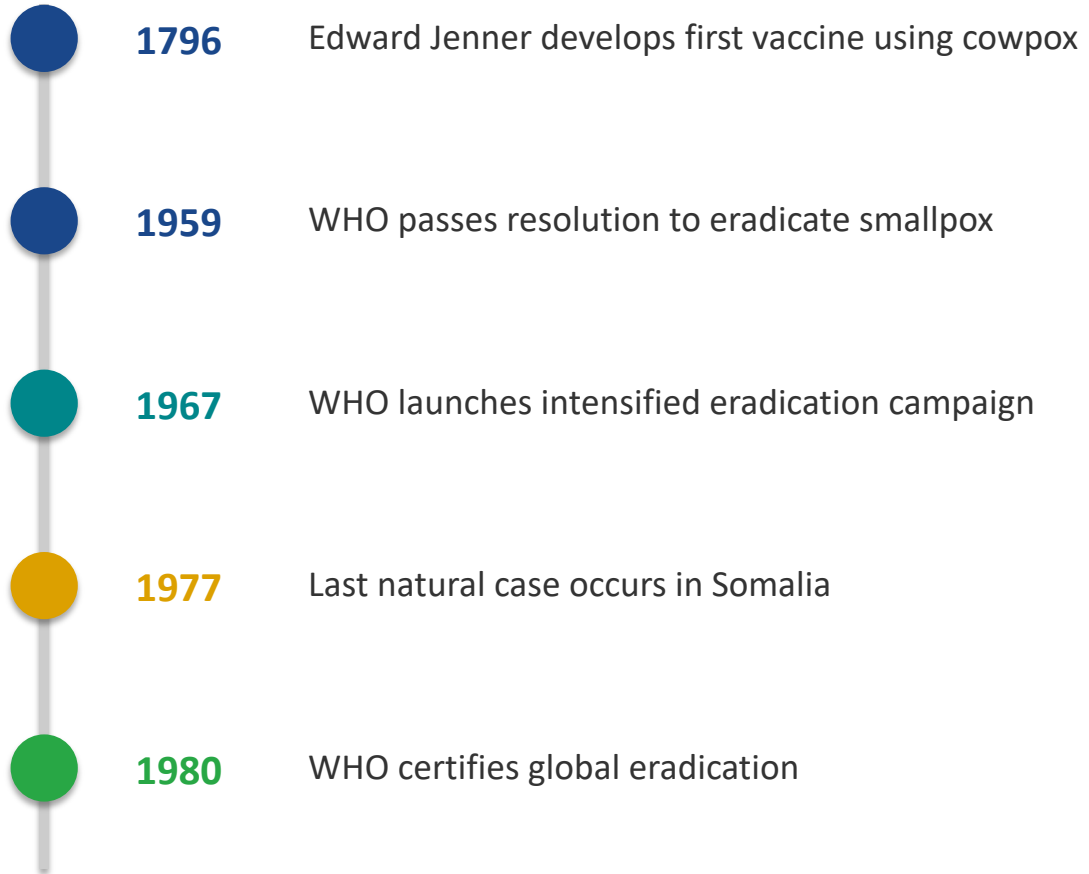
1981-Present: Post-Mandate

Cases near zero across all states; sporadic outbreaks only

Result: Before 1963, nearly every state had 100–1,000+ measles cases per 100,000. After the 1980 mandate, cases dropped to near zero across all 50 states.

Data: CDC via Our World in Data. Heatmap represents state-level incidence rates 1929–2022.

Smallpox: The Only Human Disease Eradicated Through Vaccination



Why Smallpox Could Be Eradicated:

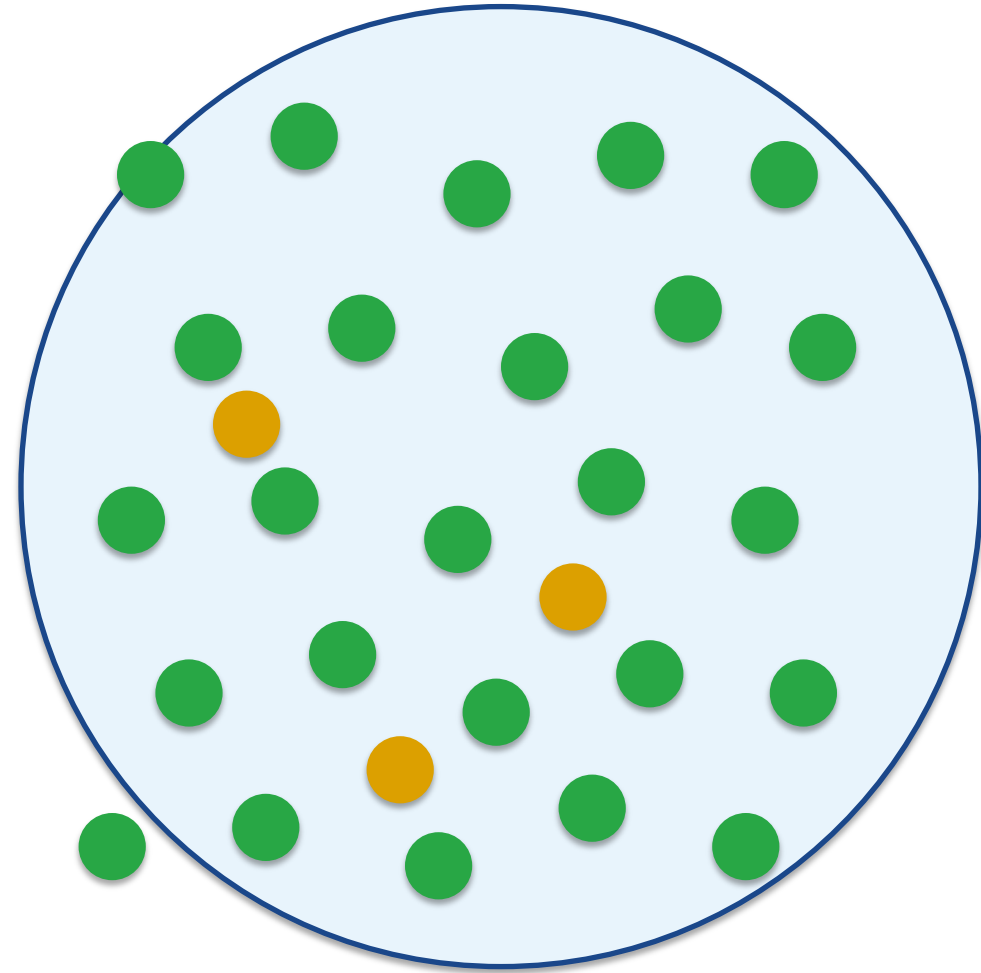
- ✓ Virus infected only humans (no animal reservoir)
- ✓ Symptoms were distinctive and visible
- ✓ Vaccine was safe, effective, and inexpensive
- ✓ Natural immunity was complete after infection

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When a sufficient proportion of a population is vaccinated, pathogen spread is blocked — protecting even those who cannot be vaccinated.

Who benefits from herd immunity?

- Newborns (too young for vaccines)
- Immunocompromised individuals
- Cancer patients on chemotherapy
- Those with vaccine allergies



- Vaccinated individuals
- Unvaccinated (protected by herd immunity)

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

40.5%

Case Reduction
Pneumococcal Disease

31.3%

Death Reduction
Pneumococcal Disease

89%

Case Reduction
Hepatitis A

83.9%

Case Reduction
Acute Hepatitis B

Why Do These Gaps Exist?

- Limited Access: Vaccines not available or affordable in all regions
- Supply Shortages: Production capacity cannot meet global demand
- Trust Deficits: Misinformation reduces vaccine uptake in some communities
- Disease Complexity: Pneumococcal has many strains; newer vaccines needed

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

100% Elimination Achieved

Diphtheria, smallpox, and both forms of polio eliminated in the US through vaccination

Development timelines collapsed

From 105 years (typhoid) to under 1 year (COVID-19) — driven by mRNA and genomic tools

Measles eliminated nationwide

Cases dropped to near zero across all 50 states following the 1980 kindergarten mandate

Smallpox — the only human disease ever eradicated

Global eradication certified in 1980 after worldwide vaccination campaign

Coverage gaps remain

Pneumococcal (40.5%), Hepatitis A (89%), Hepatitis B (83.9%) — need improved access and newer vaccines

Call to Action: Improve vaccine access, coverage, and public trust to complete the work of disease elimination.